The Power Hinge

One rate equation, three different pathways of energy transfer

Working rule: Power measures how quickly energy crosses a boundary or changes store. Begin with $P = dE/dt$, identify the particular transfer, and then add the interaction law, constitutive relation, efficiency or steady-state condition that makes the rate calculable.

Name: ____________________

Class: ____________________

Companion task to *The Placeholder Force*

1. Mechanical power: which force does the work?

An electric winch raises a 75 kg crate vertically at a constant speed of $0.80\,\mathrm{m\,s^{-1}}$. Take $g = 9.8\,\mathrm{m\,s^{-2}}$. The winch is 80% efficient.

A. Draw the transfer

Draw the crate, its upward velocity, the tension and its weight. Choose first the crate alone as the system, and then the crate–Earth system. Mark the direction of energy transfer across each boundary.

B. Apply the hinge

Constant speed gives $F_{\text{net}} = 0$ and hence $T = mg$. Calculate the positive power transferred by the tension,

$$P_T = \mathbf{T}\cdot\mathbf{v} = Tv.$$

Also calculate the power delivered by gravity and the net power on the crate. Explain why using $P = F_{\text{net}}v$ would answer a different question.

C. Add efficiency and time

Find the electrical input power to the winch. In 15 s, calculate: (i) the useful energy transferred to the crate–Earth system; (ii) the electrical energy supplied; and (iii) the energy dissipated by the winch.

D. Interrogate the placeholder

In $P = \mathbf{F}\cdot\mathbf{v}$, what exactly is $\mathbf{F}$? When can a non-zero force transfer zero power? Why is power associated with a particular interaction rather than automatically with the net force?

2. Resistive heating: one power, two substitutions

A $6.0\,\Omega$ ohmic resistor is connected directly across an ideal 12.0 V supply. Assume its resistance remains constant.

A. Draw the pathway

Sketch the source and resistor. Draw an energy-flow diagram from the source to the resistor and surroundings. Identify the meanings of current I, potential difference V and electrical power P.

B. Apply rate and closure

Use the general electrical rate relation

$$P = IV$$

and the component relation $V = IR$ to find the current and the power transferred to internal energy. State which equation describes a rate of transfer and which supplies empirical closure for this component.

C. Change what is controlled

Derive

$$P = I^2R \qquad \text{and} \qquad P = \frac{V^2}{R}.$$

Replace the resistor by $12\,\Omega$. Find the power (i) when the voltage remains 12 V and (ii) when the current is instead held at 2.0 A. Explain why increasing R can either decrease or increase P.

D. Interrogate the formula

Does I^2R itself say what the source holds fixed? Which assumptions are hidden in using a constant R? What changes if the resistor heats enough for $R = R(T)$?

3. Bulb luminosity: input, radiation and loss

A 12.0 V lamp draws 0.50 A. At its steady operating temperature, 12% of the electrical input emerges as visible radiation and 68% as infrared radiation. The rest leaves by conduction and convection. Assume its radiation spreads uniformly in all directions.

A. Partition the output

Draw a system boundary around the lamp. Show electrical power entering and the three outgoing pathways: visible radiation, infrared radiation, and conduction/convection. Add an internal-energy store inside the boundary.

B. Apply the steady-state ledger

Calculate $P_{\text{in}} = IV$. At steady state, $dU/dt = 0$, so write and complete

$$P_{\text{in}} = L_{\text{vis}} + L_{\text{IR}} + P_{\text{other}}.$$

Find each term and the total radiant luminosity $L_{\text{rad}} = L_{\text{vis}} + L_{\text{IR}}$.

C. From luminosity to intensity

At $r = 2.0\,\mathrm{m}$, calculate the total radiant intensity

$$\mathcal{I} = \frac{L_{\text{rad}}}{4\pi r^2}$$

and the visible radiant intensity. Distinguish electrical wattage, radiant luminosity, visible radiant power and intensity at a detector.

D. Before steady state

Immediately after switch-on, why is $dU/dt > 0$? Complete $P_{\text{in}} = dU/dt + P_{\text{out}}$. Why does a hot filament require a temperature-dependent constitutive relation rather than one fixed resistance?

Synthesis: power gives the rate; the model supplies the pathway

1. Complete the comparison.

Case	General rate hinge	Extra relation or condition	What does the power describe?
Winch			
Resistor			
Lamp			

2. Develop the central claim.
Respond to:

"Power is a rate, not a mechanism."

Use all three cases. Include the terms *system boundary*, *interaction-specific force*, *constitutive relation*, *controlled condition*, *efficiency*, and *steady state*. Explain why $P = \mathbf{F}\cdot\mathbf{v}$, $P = IV$, $P = I^2R$ and luminosity in watts can all measure rates of energy transfer while describing different physical pathways.

Teacher guide and indicative answers

The Power Hinge

1. Mechanical power

Forces and rate. Constant speed means

$$T = mg = 75(9.8) = 735\,\mathrm{N}.$$

The power delivered by the tension is

$$P_T = Tv = 735(0.80) = 588\,\mathrm{W}.$$

Gravity delivers

$$P_g = -mgv = -588\,\mathrm{W},$$

so the net power on the crate is zero, consistent with constant kinetic energy.

Why the useful power is non-zero. $P = \mathbf{F} \cdot \mathbf{v}$ refers to a specified force. The winch transfers energy at 588 W even though the net force and net power on the crate are zero. For the crate–Earth system, that input increases gravitational potential energy.

With efficiency $\eta = 0.80$,

$$P_{\mathrm{in}} = \frac{P_{\mathrm{useful}}}{\eta} = 735\,\mathrm{W}.$$

In 15 s,

$$E_{\mathrm{useful}} = 588(15) = 8.82\,\mathrm{kJ},$$
$$E_{\mathrm{in}} = 735(15) = 11.025\,\mathrm{kJ},$$
$$E_{\mathrm{diss}} = 2.205\,\mathrm{kJ}.$$

Conceptual answer. A force perpendicular to velocity transfers zero instantaneous power. The dot product identifies both the force and the direction of motion. Net force governs momentum change; the power of an individual interaction governs its contribution to energy transfer.

Diagnostic misconception. "Constant speed means no work is being done." It means zero net work on the crate, while positive and negative powers may cancel.

2. Resistive heating

Ohm's law gives

$$I = \frac{V}{R} = \frac{12.0}{6.0} = 2.0\,\mathrm{A}.$$

The transfer rate is

$$P = IV = 2.0(12.0) = 24\,\mathrm{W}.$$

Using $V = IR$,

$$P = I(IR) = I^2R,$$

and using $I = V/R$,

$$P = \frac{V}{R}V = \frac{V^2}{R}.$$

For $R = 12\,\Omega$ at fixed $V = 12\,\mathrm{V}$,

$$P = \frac{12^2}{12} = 12\,\mathrm{W}.$$

At fixed $I = 2.0\,\mathrm{A}$,

$$P = I^2R = 2.0^2(12) = 48\,\mathrm{W}.$$

Conceptual answer. There is no contradiction. At fixed voltage, increasing resistance reduces current strongly enough to reduce power. At fixed current, the source raises the voltage needed to drive that current, so power increases. The algebraic form does not state the controlled condition. $P = IV$ is the general electrical energy-transfer rate. $V = IR$ is empirical closure for an ohmic component in a regime where R may be treated as constant. If temperature changes substantially, $R = R(T)$ and electrical heating couples to a thermal model.

Diagnostic misconception. Treating I^2R as a free-standing causal law while ignoring how I, V and R are constrained by the source and component.

3. Bulb luminosity

The electrical input is

$$P_{\mathrm{in}} = IV = 12.0(0.50) = 6.0\,\mathrm{W}.$$

At steady state,

$$\frac{dU}{dt} = 0$$

and therefore

$$6.0 = L_{\mathrm{vis}} + L_{\mathrm{IR}} + P_{\mathrm{other}}.$$

The pathways are

$$L_{\mathrm{vis}} = 0.12(6.0) = 0.72\,\mathrm{W},$$
$$L_{\mathrm{IR}} = 0.68(6.0) = 4.08\,\mathrm{W},$$
$$P_{\mathrm{other}} = 0.20(6.0) = 1.20\,\mathrm{W}.$$

Thus

$$L_{\mathrm{rad}} = 0.72 + 4.08 = 4.80\,\mathrm{W}.$$

At $r = 2.0\,\mathrm{m}$,

$$\mathcal{I}_{\mathrm{rad}} = \frac{4.80}{4\pi(2.0)^2} \approx 9.55 \times 10^{-2}\,\mathrm{W\,m^{-2}},$$
$$\mathcal{I}_{\mathrm{vis}} = \frac{0.72}{4\pi(2.0)^2} \approx 1.43 \times 10^{-2}\,\mathrm{W\,m^{-2}}.$$

Conceptual answer. The 6 W rating is electrical input. Radiant luminosity is total emitted radiant power; visible radiant power is only one spectral part; intensity is power received per unit area at a location. Photometric luminous flux in lumens would add the eye's wavelength sensitivity and is not calculated here.

During warm-up,

$$P_{\mathrm{in}} = \frac{dU}{dt} + P_{\mathrm{out}}, \qquad \frac{dU}{dt} > 0.$$

A filament's resistance rises strongly with temperature, so a complete model needs $R(T)$ together with radiation, conduction and convection.

Synthesis and classroom use

Indicative synthesis.

All three cases begin with power as a rate of energy transfer, but the symbol does not identify the pathway. In the winch, $\mathbf{F} \cdot \mathbf{v}$ is the power of a specified interaction, and positive tension power can coexist with zero net power. In the resistor, $P = IV$ becomes I^2R only after the ohmic constitutive relation is inserted; whether power rises or falls with resistance depends on the controlled condition. In the lamp, electrical input divides among internal-energy change, radiation and other thermal transfers. Steady state removes dU/dt, while efficiency and spectral fractions determine how much input becomes luminosity. The rate equation constrains the account; the physical model names the pathway.

Discussion prompts.

- Which force belongs in $\mathbf{F} \cdot \mathbf{v}$, and why need it not be the net force?
- Which equation is the general rate relation and which equation closes the resistor model?
- What must be held fixed before a claim about how P varies with R has meaning?
- Why is a bulb's electrical wattage not automatically its visible luminosity?
- What term vanishes at steady state, and what physical process is hidden when it is removed?

Suggested timing: 9 minutes per case, 10 minutes synthesis, 8 minutes whole-class comparison.